Random Walks

ADVANCED TOPICS IN NETWORK SCIENCE · MODULE 07

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#1 title

Where is the walker after 100 steps?

A man walks 1 yard, turns in a random direction, and walks 1 yard again. He repeats this 100 times.

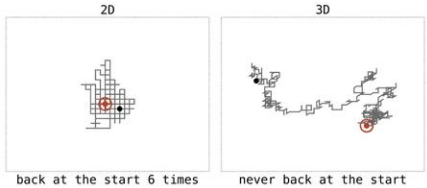
Pearson asked this in *Nature* in 1905.



Far from the start, or near it? Tell your neighbour.

Pub: Jim Oxley, CC BY-SA 2.0 (geograph.org.uk)

#2 hook (story): Pearson's question, 19



2D: the walker always comes home. 3D: only 34% of walkers ever do.

#3 story: Polya's recurrence (2D return

Random walks appear in many fields



Mosquito: CDC / James D. Gathany (public domain) · trajectory: Jean Perrin (public domain)

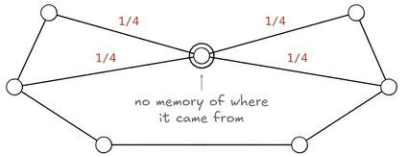
#4 hook: the same walk in three fields

How does it tell us about a network?

But it is a "random" walk, right?

#5 spine 1: the question

What is a random walk?



#6 mechanism: the rule

Pen & Paper

10 minutes · stop at the line on the sheet

#7 Pen & Paper: first block

Random walk simulator

skojaku.github.io/
adv-net-sci/assets/vis/
random-walks/index.html

#8 the simulator (QR code and link)

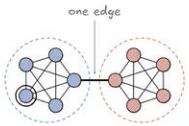
Two communities joined by one edge

How many steps, on average, until the walker crosses to the other community?

A: about 3

B: about 20

C: about 200

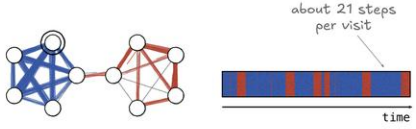


Take a guess.

#9 question to the room (Q2)

A short walk stays in its community

OPTIONAL



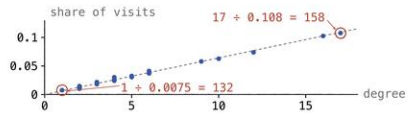
#10 answer to Q2

Short walks show communities. Long walks show degree. Why?

#11 spine 2: the mystery

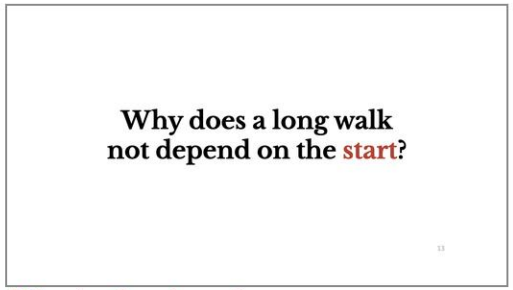
Degree ÷ share is constant

Karate club: 34 nodes, 78 edges, so 2m = 156

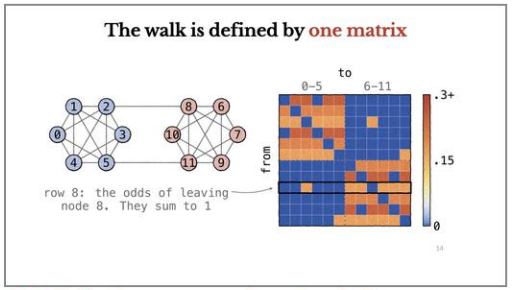


Share of time = degree ÷ 2m

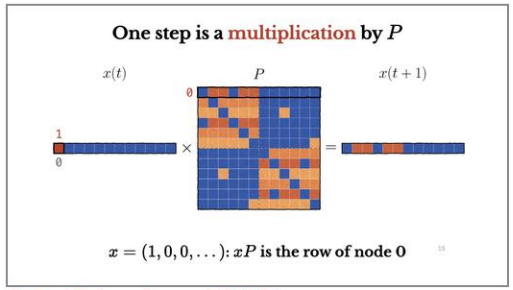
#12 debrief Part 1 (degree ÷ share)



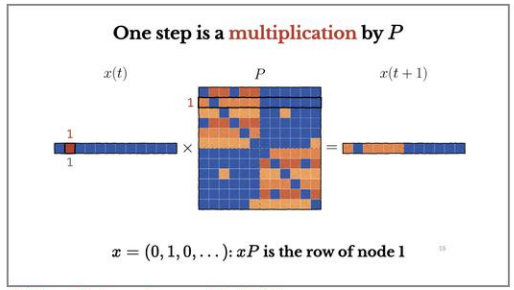
#13 spine 3: part question



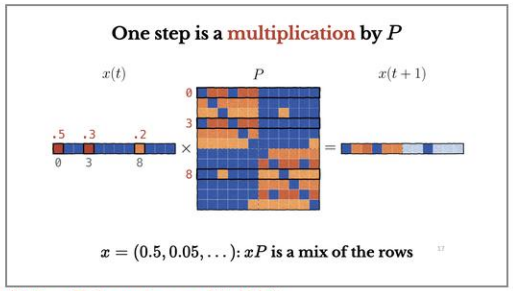
#14 P (the two-community network, the m



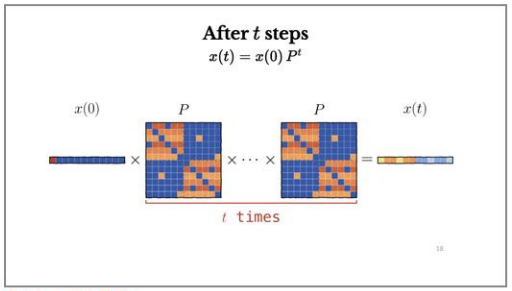
#15 x P, how to read it (1/3)



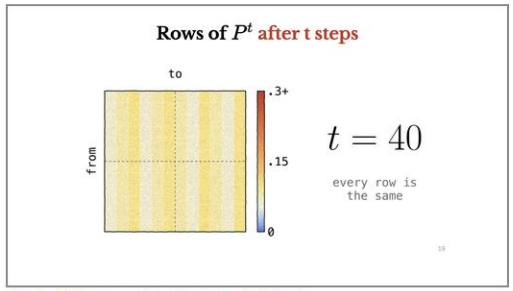
#16 x P, how to read it (2/3)



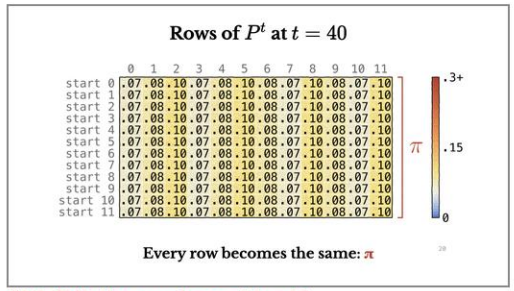
#17 x P, how to read it (3/3)



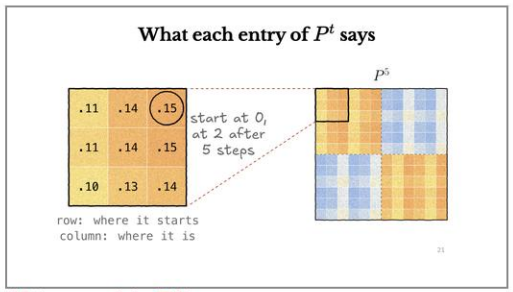
#18 x(0) P^t



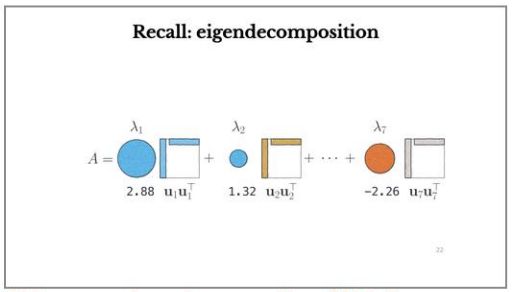
#19 P^t as a heat map (slider)



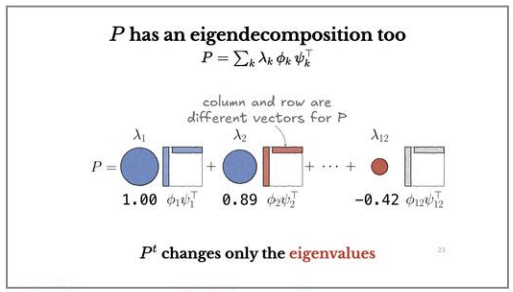
#20 P^t, the numbers at t = 40



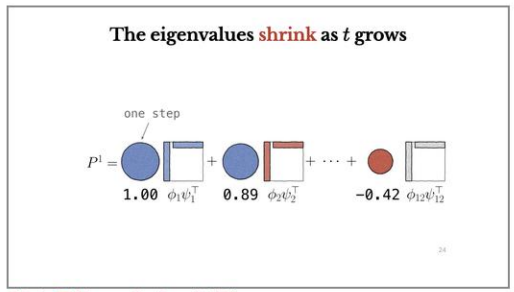
#21 zoom into P^t



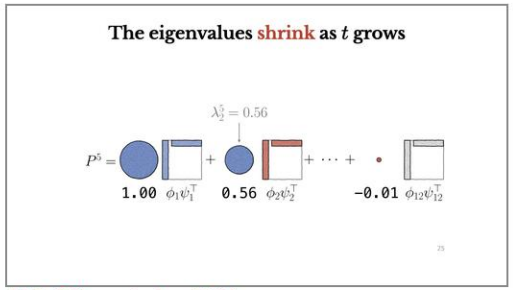
#22 recap: eigendecomposition (M06, lig



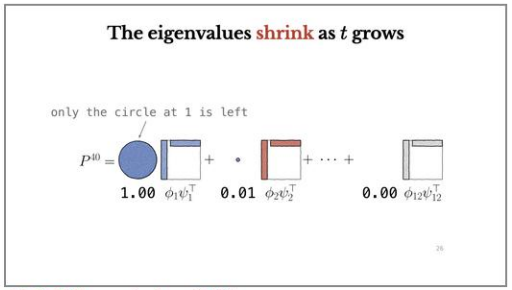
#23 P as a sum of circles and bars



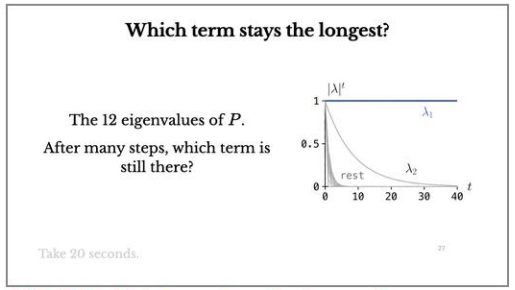
#24 P^t as circles (1/3)



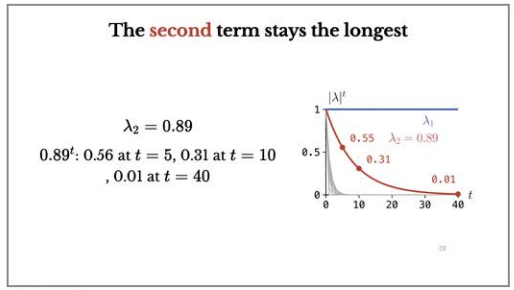
#25 P^t as circles (2/3)



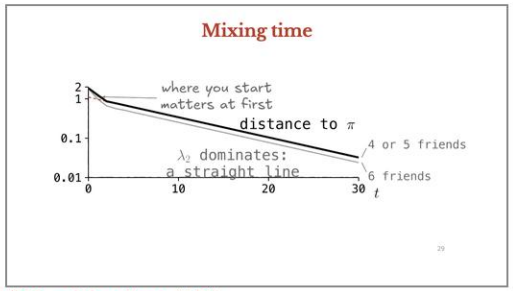
#26 P^t as circles (3/3)



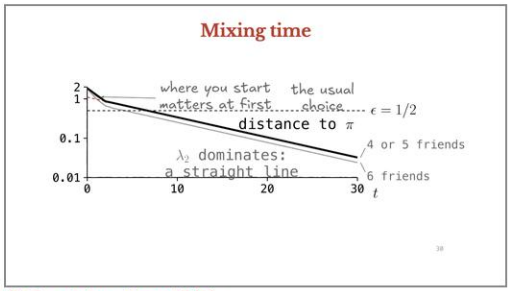
#27 Q3 (which term stays the longest)



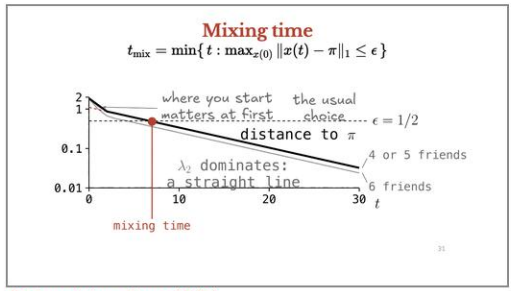
#28 A3



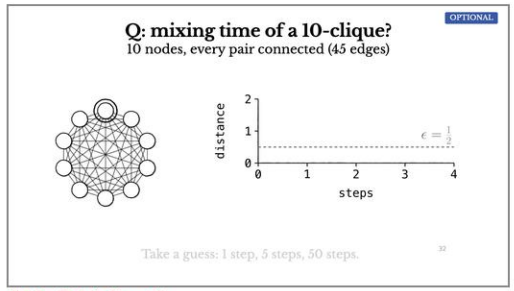
#29 mixing time (1/3)



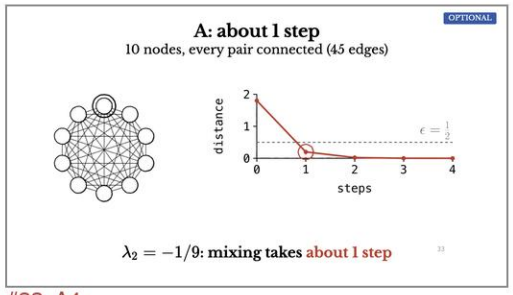
#30 mixing time (2/3)



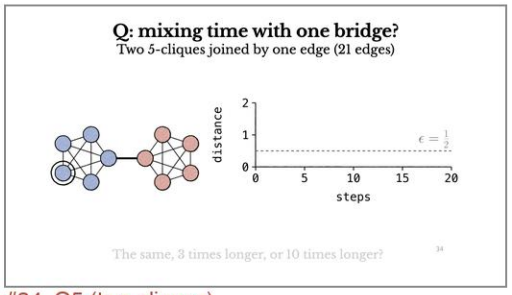
#31 mixing time (3/3)



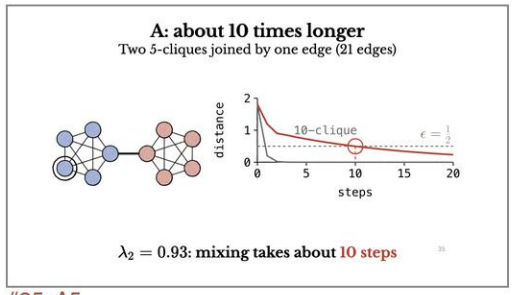
#32 Q4 (clique)



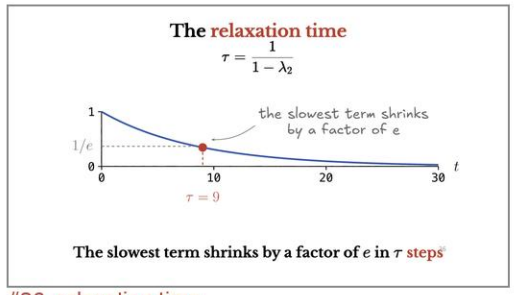
#33 A4



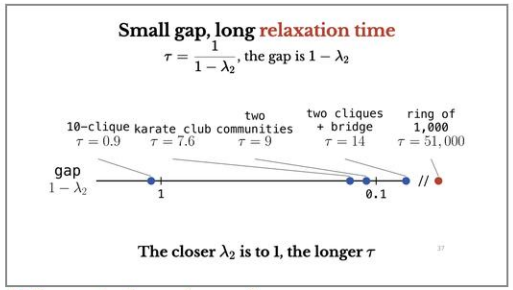
#34 Q5 (two cliques)



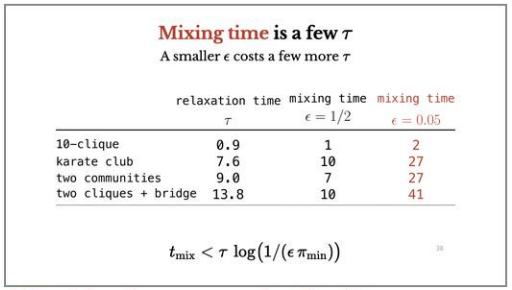
#35 A5



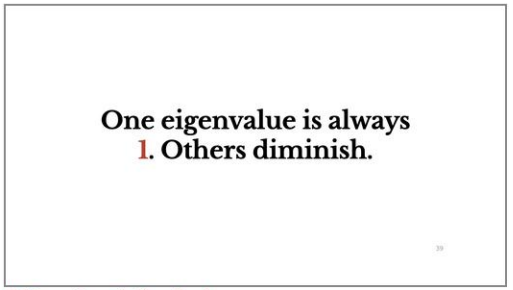
#36 relaxation time



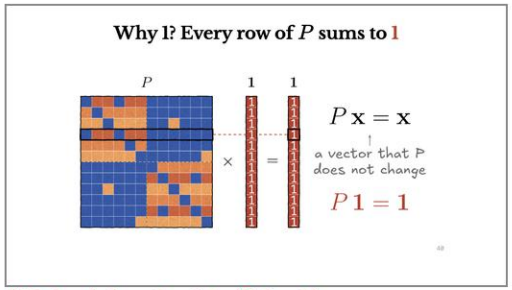
#37 spectral gap (gauge)



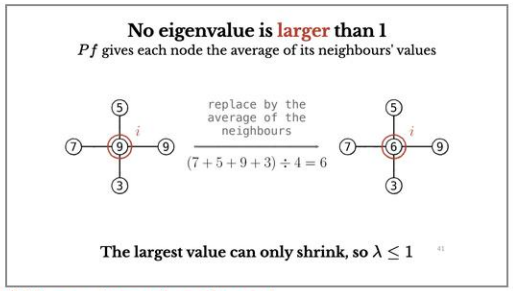
#38 mixing time versus relaxation time



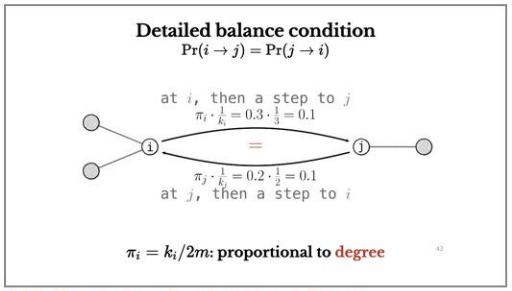
#39 spine 4: the fact



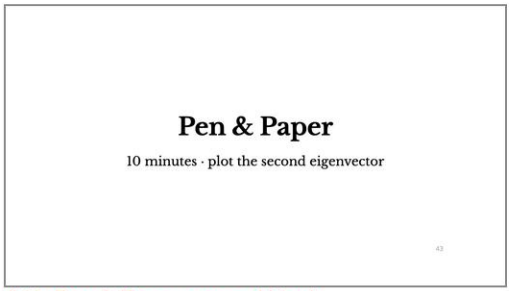
#40 lambda = 1 exists (P 1 = 1)



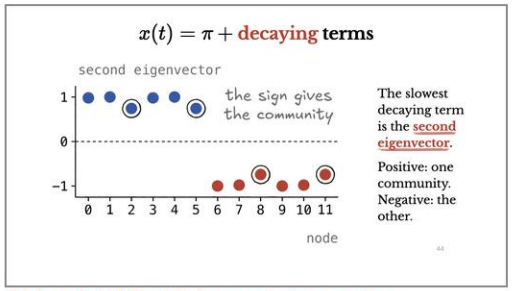
#41 no eigenvalue above 1



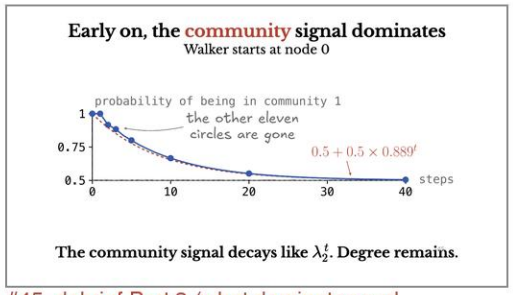
#42 the eigenvalue 1 gives pi (ties to



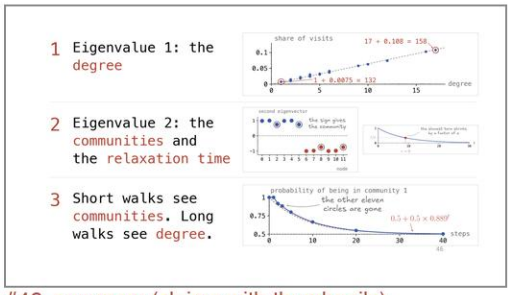
#43 Pen & Paper: second block



#44 debrief Part 3 (second eigenvector;



#45 debrief Part 3 (what dominates earl



#46 summary (claims with thumbnails)